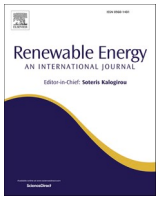
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Enhancing methanogenesis efficiency in anaerobic digestion of food waste through incremental salinity and nanobubble utilization to domesticate inoculum: Adaptability and optimal inoculation ratios

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ABSTRACT

High salinity inhibits methane production in food waste (FW) anaerobic digestion (AD). This study investigated the salinity tolerance (1 %–3 %) of inoculum acclimated with air nanobubble water (Air-NBW) and evaluated methanogenic potential under 3 % salinity across varying inoculum-to-substrate (I/S) ratios (volatile solids (VS) basis). As results, the methane yield of the Air-NBW group significantly increased by 31 %, 18 %, and 16 % ($p < 0.05$ for all phases) relative to the control group across the three phases. Remarkably, the lag period in the Air-NBW group exhibited a notable reduction as the salinity increased, particularly at the salinity of 2 % where a significant decrease of 14 days in the lag period was observed in comparison to the control group. Additionally, the Air-NBW digester showed a non-significant increase in methane yield (228.88 ± 4.36 mL/g-VS_{fed}, $p = 0.061$) at I/S = 1, while demonstrating a significant increase to 307.45 ± 3.77 mL/g-VS_{fed} at I/S = 0.5 ($p = 0.007 < 0.05$) compared to the DW groups at equivalent I/S ratios. These results indicate that the using a salinity-adaptable inoculum could effectively facilitate the swift initiation and sustained stability of FW digesters.

1. Introduction

The rapid growth of the catering business has generated considerable Food waste (FW) that should be urgently disposed of in an effective manner. Recently, FW has been confirmed as one of the preferred substrates for anaerobic digestion (AD) for energy recovery since it covers high levels of biodegradables (e.g., polysaccharides, proteins, and oils) [1]. However, several environmental factors have frequently hindered the AD of FW. For instance, high salinity in FW, a common challenge, is primarily attributed to the extensive use of sodium chloride-based salt (typically > 97 % in China) as a cooking seasoning [2].

For the AD system, Na^+ exerts more significant inhibiting effects than Cl^- [3]. A specific limit of Na^+ concentration is favorable for the metabolic activity of microorganisms (e.g., facilitating the synthesis of adenosine triphosphate, or stimulating the oxidation of nitramine adenine dinucleotides) [4,5]. Conversely, high Na^+ concentration will be toxic to microbial metabolism to a certain extent. When external osmotic pressure exceeds intracellular pressure, water flows out, causing cellular protoplasm separation, leading to cell lysis and death [6]. Also, high salinity is likely to result in lower dehydrogenase activity, serving as a vital indicator of microbial activity in the AD digesters [7]. Existing research has confirmed that a variety of salinity levels produced varying inhibition degrees of CH_4 yield in the AD of FW digesters, with higher inhibition levels with the increase of the salt levels. For instance, the inhibition degrees of CH_4 yield in the one-phase AD of kitchen waste system with 10–16 g NaCl/L range from 24 % to 81 % [8]. Zhang et al. [9] suggested that high salinity (>15 g/L)) would reduce the diversity and the abundance of methanogens. Besides, NaCl concentration also

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affects the biogas composition from kitchen waste AD, with methane content decreasing as NaCl concentration increases [10]. Therefore, relieving the inhibition of high salinity is essential to improve methane production from AD of FW.

The existing common strategy refers to the reduction of the salinity in the AD of the FW system by co-digestion with other low-salt substrates or by using water and wastewater to dilute the salinity of the AD system. A further strategy is to resist the adverse high-salt environment by improving microbial activity and salt tolerance of microorganisms. External osmoprotectants (e.g., betaine, trehalose, and amino acids) can effectively restore the activity of methanogens under high-salt conditions [11]. Additionally, existing research has confirmed that undomesticated anaerobic microorganisms took nearly 20 d to adapt to high-salt environments [12]. In contrast, anaerobic microorganisms that have been acclimated to salt could effectively counteract the inhibitory influences of certain salinity levels on methanogenesis [13]. As indicated by the Zhang et al. [10], although methane production, dehydrogenase, and coenzyme F_{420} activities were first enhanced and then decreased with increasing salt concentration, the methanogenic bacteria (*Methanosaeta* and *Methanosarcina*) exhibited potential adaptability to high-salt environments after domestication.

Nanobubble water (NBW) refers to water containing nanobubbles smaller than 1 μm in diameter, noted for its pollution-free nature, high stability, superior gas solubility, and negative zeta potential. Previous research has confirmed that NBW possesses the remarkable ability to significantly increase microbial activity [12,14,15]. Hence, the introduction of NBW into AD systems has emerged as a novel strategy, proven to boost the bioactivities of enzymes and microorganisms that are vital in the AD processes [16,17]. Most recently, a novel investigation has exploited NBW technology for alleviating the inhibitory effect of high-salt concentration on methanogenic archaea, such that excellent results of 14 %–43 % increase in methane yields and 26 %–58 % increase in coenzyme F_{420} activity have been achieved [12]. As revealed by the above-mentioned findings, NBW can serve as an effective strategy for addressing the adverse effects of high-salt levels in the AD system. Specifically, by utilizing NBW to domesticate AD sludge as inoculum, the efficiency of hydrogen production in two-stage AD of FW system under salt stress conditions of 20 g NaCl/L can be markedly enhanced, and the start-up time of the digester can be substantially shortened [18]. Therefore, adopting a strategy that combines NBW with a gradual increase in salinity for domesticating methanogens may represent an economical and environmentally friendly approach aimed at optimizing the AD efficiency of high-salinity FW. Furthermore, the identification of an optimal inoculation ratio (inoculum/substrate (I/S)) in the AD system has been heavily investigated since it is a critical parameter that exerts a certain effect on the stability of the AD digesters [19,20]. However, the optimal I/S ratios for AD of organic solid wastes in the adverse environment have been rarely investigated, such that high-salinity FW was used in this study.

In this study, the adaptability of inoculum domesticated by NBW in the AD of FW was investigated with the increasing salinity levels ranging from 1 % to 3 %, to confirm the optimal I/S ratio for AD of the high-salinity FW system. First, an acclimatization process was performed by ever-increasing salinity from 1 % to 3 %, and the Air-NBW was introduced in the first phase (salinity was 1 %) to domesticate the inoculum. The samples that were taken at the end of the respective phase were analyzed to determine the electron transport system (ETS) activity, and the result can reveal the microbial activity in response to increased salinity during the AD of FW with or without the addition of Air-NBW domesticated inoculum. Afterward, batch tests were performed at varying I/S ratios in the AD of FW under a salinity of 3 % using the domesticate inoculum to investigate the effects of inoculation ratios on methane production, as well as the concentration of volatile fatty acids (VFAs) and alkalinity. This study aimed to provide a long-term stability strategy for AD of the high-salinity FW process.

2. Materials and methods

2.1. Inoculum, FW, and Air-NBW

In this study, anaerobically digested sludge (AD sludge) was utilized as the inoculum and activated under mesophilic conditions (38 ± 1 °C) for approximately two weeks, until gas production ceased. The initial total solids (TS) and volatile solids (VS) of the inoculum were 2.45 % ± 0.02 % and 1.92 % ± 0.02 %, respectively. A simulated FW with a carbon-to-nitrogen (C/N) ratio of 18.28, comprising noodles (22 %), rice (25 %), pork (16 %), cabbage (29 %), and tofu (8 %), was employed in this study, and its TS and VS were determined to be 18.01 % ± 0.01 % and 17.59 % ± 0.03 %, respectively. Additionally, the total VFAs concentration and pH value were measured as 904.49 ± 24.20 mg/L and 5.72 ± 0.03, respectively. Consistent with the methodology described in Ref. [18], the Air-NBW was manufactured using a gas-water circulation method facilitated by an NBW generator. The apparatus was fed with both water and air through piping systems, where the flow rates were meticulously regulated at 0.05 L/min for the gas and 3 L/min for the water. Throughout the entire process, a constant pressure of 0.25 MPa was maintained to ensure optimal conditions for the formation of nanobubbles. Notably, the air utilized for the generation of Air-NBW was sourced directly from the natural environment of the laboratory. The Air-NBW with a bubble concentration of $(2.9 \pm 0.3) \times 10^8$ particles/mL and a particle size of 38–115 nm was determined after 20 min of preparation.

2.2. Acclimatization process by addition of Air-NBW and gradually increasing NaCl concentration from 1 % to 3 %

The acclimatization process fell into three phases with the progressively increasing salt concentration. In the first phase, the inoculum was domesticated in the AD of FW (salinity of 1 %) system with or without Air-NBW introduced, which were termed deionized water (DW) (as the control group) and Air-NBW. The experiments were performed in the 500 mL digesters with the addition of 200 g AD sludge and 10 g FW (wet weight) with the I/S ratio (VS basis) of 2 to enrich halotolerant microbial consortia by alleviating substrate overload, and the working volume was regulated to 350 mL by the DW or Air-NBW. The initial pH reached 6.82 ± 0.09, and not any buffering agents were added to adjust the pH value. Subsequently, the NaCl concentration was increased to 2 % and 3 % in the second and third phases, respectively. For the above-mentioned two phases, there was no re-addition of Air-NBW, but the sludge domesticated by DW or Air-NBW in the first phase served as inoculum. The respective digester was fed with the I/S ratio of 2, and the pH reached 7.27 ± 0.05 and 6.99 ± 0.05 in the second and third phases, respectively. Lastly, the samples were taken at the end of the respective phase to analyze the ETS activity.

2.3. Different I/S ratio tests in the AD of FW under a salt level of 3 % by using the domesticate inoculum

After the acclimatization process, the I/S ratio continuously declined from 2 to 1 and 0.5 for feeding FW to confirm the appropriate I/S for the high salinity (3 %) FW AD system. The initial pH of all digesters was regulated to 7.5 using 2 M HCl and 2 M NaOH solution. In the above process, the samples were regularly taken to determine the variations in volatile fatty acids (VFAs) and alkalinity (e.g., the total alkalinity (TA), partial alkalinity (PA), as well as intermediate alkalinity (IA)).

For all experiments, the hermetically sealed digesters were purged using pure N_2 about 3 min, 3 times prior to initiation of the experiments to ensure that the anaerobic condition persisted and then operated in an incubator with a temperature of 38 ± 1 °C.

2.4. Analytical and calculational methods

All liquid samples were first centrifuged to produce the supernatant, and then the supernatant was filtrated with a 0.45 μm filter. The standard methods were adopted to determine the common parameters (e.g., TS, VS, TA, PA, and IA) [21]. The concentrations of six common VFAs, including acetic acid (HAc), propionic acid (HPr), iso-butyric acid (iso-HBu), n-butyric acid (n-HBu), isovaleric acid (iso-HVa), or n-valeric acid (n-HVa), were analyzed using a gas chromatograph armed with a Unisole F-200 30/60 column and flame ionization detector. In this study, the ETS activities were determined by the iodonitrotetrazolium dehydrogenase (INT-ETS method) [22]. To be specific, 0.5 mL of sample homogenate was taken from the end of the respective phase with a syringe and injected directly into a 10 mL centrifuge tube. Next, 0.1 mL of INT solution (0.2 %, w/v) was introduced. The samples were incubated in a 37 ± 1 °C table concentrator by shaking at 100 rpm/min for 30 min according to the standard, followed by the addition of 1 mL of methanol (37 %). After the incubation was terminated, the sample was centrifuged at 10,000 r/min for 5 min at 4 °C, and the precipitate was introduced into 5 mL of methanol. Subsequently, the respective tube was shaken in the dark for 10 min at 37 ± 1 °C and was centrifuged at 10,000 rpm/min for 5 min to take the supernatant to measure absorbance under 485 nm (A_{485}). The sediment gained after centrifugation was dried to a constant weight in an oven at 105 °C, and then the sample was weighed. The INT-ETS activity can be determined by Eq. (1) [14]. The pH value was monitored using a pH meter (Testo 206, Germany), and all the measurements were performed in triplicate.

$$\text{INT-ETS activity} = A_{485}\cdot V/(K_T\cdot W\cdot T) \quad (1)$$

where the unit of INT-ETS activity is mg/(g·h); A_{485} denotes absorbance under 485 nm; V represents methanol volume; K_T expresses the slope of the standard curve; W denotes the dry weight of the precipitate; T is reaction time (0.5 h).

For the measurement of gas samples, the biogas volumes were examined using 20- and 50-mL air-tight syringes at 25 °C and 1 atm. Subsequently, the contents of CH_4 and CO_2 were determined using another gas chromatograph (GC-8A, Shimadzu, Japan).

2.5. Data processing and statistical analysis

All experiments were performed in duplicate, with each sample measured in triplicate. The data were analyzed using Microsoft Office Excel 2010 and are presented as mean ± standard deviation. One-way analysis of variance (ANOVA) was conducted to determine statistical differences among groups using IBM SPSS Statistics, with a p-value <0.05 considered statistically significant. All graphical representations were prepared using Origin 2019.

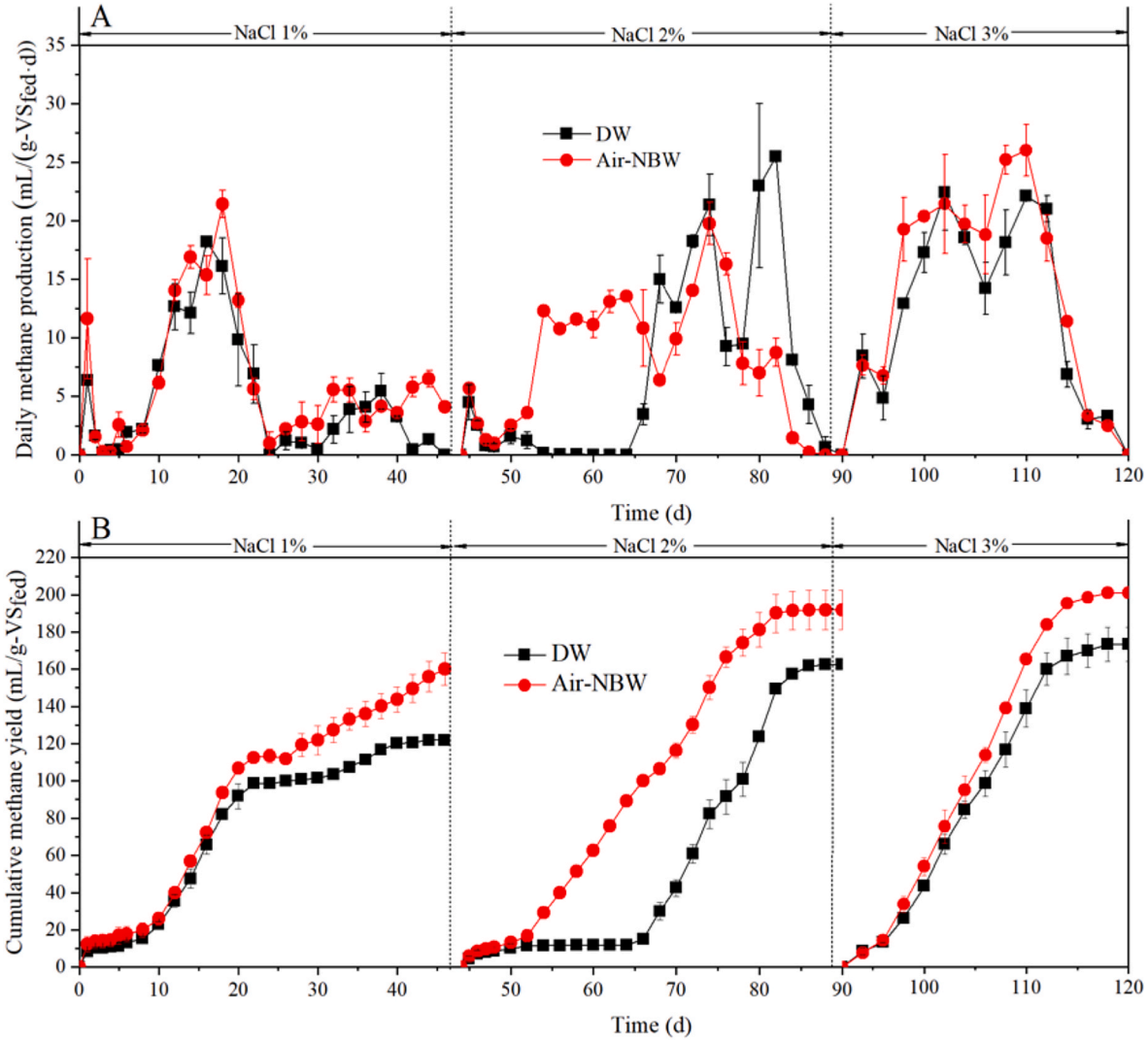


Fig. 1. Daily methane production (A) and cumulative methane yield (B) during the acclimatization process with increasing salt concentration.

3. Results and discussion

3.1. AD performance of the acclimatization process and ETS activity

3.1.1. AD performance of FW with different salinity during the acclimatization process

In the first phase, no substantial difference was reported in the daily methane production between the DW and Air-NBW digesters (Fig. 1 A), and the maximum methane production rate reached 18.22 ± 0.12 and 21.42 ± 1.15 mL/(g-VS_{fed}·d) on 16 days and 18 days in the DW and Air-NBW digester at the salinity of 1 %, respectively. Notably, a long start-up time of approximately 22 d was identified in the DW group with the salinity increased to 2 %, whereas none of this happened in the Air-NBW group. As reported by Hou et al. [12], there was a longer start-up period (approximately 20 d) when the salt stress was 20 g NaCl/L with or without adding Air-NBW in the methanogenesis phase of two-phase AD of the FW system with the I/S ratio of 0.5. This indicates that a significant reduction in the start-up period of the digesters could be achieved by domesticating the inoculum with Air-NBW added and increasing salt concentration. This is presumably due to the fact that the domestication of microorganisms using Air-NBW enhances their adaptability and resilience to stressful conditions such as high salt concentrations. By improving mass transfer, enhancing nutrient utilization, promoting biological activity, and enabling gradual adaptation, it collectively facilitates the faster establishment of a functional microbial community, thereby significantly reducing the start-up time of AD process.

In addition, the acclimatization process was first operated at the high I/S ratio (I/S = 2), such that over-acidification and other undesirable phenomena can be avoided effectively. Li et al. [20] demonstrated that the acid suppression phenomenon was problematic in the low I/S ratio in the AD of FW digesters. For the high-salinity FW, the effects of I/S ratios on the methanogenesis performance are discussed in detail in section 3.2 of this study. Furthermore, a salt adaptation was obtained through a long-term acclimatization process, suggesting that there was virtually no lag time in both groups when the salt concentration continued to be increased to 3 %. The peak values of daily methane production in this phase were elevated to 22.11 ± 0.12 and 26.021 ± 2.21 mL/(g-VS_{fed}·d) with or without the addition of the Air-NBW, respectively.

Multitudinous studies proved that high-salt concentration could dramatically influence AD performance. For instance, Zhao et al. [23] reported that the maximum methane yield was only 22.7 mL/g-VS_{fed} in the one-phase FW AD digester with the 15.0 g NaCl/L. Liu et al. [24] also found that the methane yield in the one-phase AD of kitchen waste was decreased from 311.9 to 133.6 mL/g-VS_{fed} when the salt concentration was increased from 0 to 25 g NaCl/L. In contrast, the cumulative CH_4 yields in both groups were increased rather than decreased with the increase of the salinity from 1 % to 3 % in this study (Fig. 1 B). To be more specific, the final cumulative methane yield was improved from 122.15 ± 0.63 to 173.21 ± 9.14 mL/g-VS_{fed} in the DW group ($p = 0.005 < 0.05$) and from 160.13 ± 8.67 to 201.02 ± 0.53 mL/g-VS_{fed} ($p = 0.044 < 0.05$). As expected, all cumulative methane yields in the Air-NBW digesters were more efficient than those in the DW digesters in the three phases, marking an increase of 31 %, 18 %, and 16 %, respectively, and semblable conclusions were reported by Chen et al. [25].

Furthermore, Wang et al. [15] have discovered that O_2-NBW demonstrates remarkable effectiveness in facilitating transmembrane transport of organic substances, which might accelerate the growth process of methanogenic microorganisms. The micro-oxygen environment created by O_2-NBW positively stimulates the synthesis of ATP, enabling more efficient metabolic activities and providing sufficient energy support for cellular processes such as metabolism, growth, and signal transduction. In high-salinity environments, the cellular membranes of microorganisms may be damaged, affecting the efficiency of substance transport, and frequently, their energy metabolism is severely impacted, leading to decreased growth rates and slowed metabolic activities. However, NBW could assist in improving the efficiency of substance transport for microorganisms under high-salinity conditions, stimulating more efficient metabolic activities, and thereby mitigating the inhibitory effects of high salinity to a certain extent.

3.1.2. ETS activity

Microbial activity is capable of largely regulating the outcome of methane production in AD systems, especially in the presence of undesirable factors. To a certain extent, ETS activity reveals the adaption of microorganisms in the AD system at high salinity, and Fig. 2 presents the results of ETS activity at the end of the respective phase, consistent with the results of methanogenesis performance. Moreover, the ETS activity was increased from 25.10 ± 4.69 and 35.98 ± 5.06 mg/(g·h) to 68.69 ± 8.28 and 87.03 ± 10.03 mg/(g·h) in the DW and Air-NBW digesters at the salinity elevated from 1 % to 3 %, respectively. The experimental data demonstrate that acclimating microbial inoculum through Air-NBW introduction during the initial phase could significantly enhances enzymatic activity (particularly ETS activity) in AD systems under hypersaline stress. This metabolic stimulation likely explains the superior methane production observed in Air-NBW reactors compared to conventional acclimation approaches.

As revealed by the above-mentioned results, using the inoculum domesticate through the introduction of Air-NBW in the first phase and increasing salt concentration can lead to the significantly enhanced tolerance of microorganisms under the high-salt concentration in the AD digester. Compared with our existing research, this strategy may be more effective in obtaining higher methane production. Another study by Chen et al. [25] has suggested that the improvement of microbial adaptations by the gradually increased salt concentration leads to the increased abundance ratio of halotolerant microbes, as confirmed by an enhancement in their hydrolytic and acidogenic abilities with the incremental increase in salinity. Furthermore, Zhang et al. [10] confirmed that the abundance ratios of *Methanosaeta* and *Methanosarcina* are elevated from 16.46 % and 8.77 % to 22.08 % and 10.01 %, respectively after acclimatization with the rise of the NaCl concentration in the AD of kitchen waste. Likewise, Wang et al. [26] indicated that *Methanolinea*, *Methanospirillum*, and *Methanoculleus* exhibit varying degrees of adaptation to salinity. Building on these findings, Air-NBW technology could activate microbial metabolic pathways and associated enzymatic cascades. Complementarily, the phased salinity acclimatization protocol employs incremental ionic strength adjustments, enabling strategic microbial community succession toward halotolerant phenotypes. Therefore, the integrated strategy of Air-NBW application with progressive

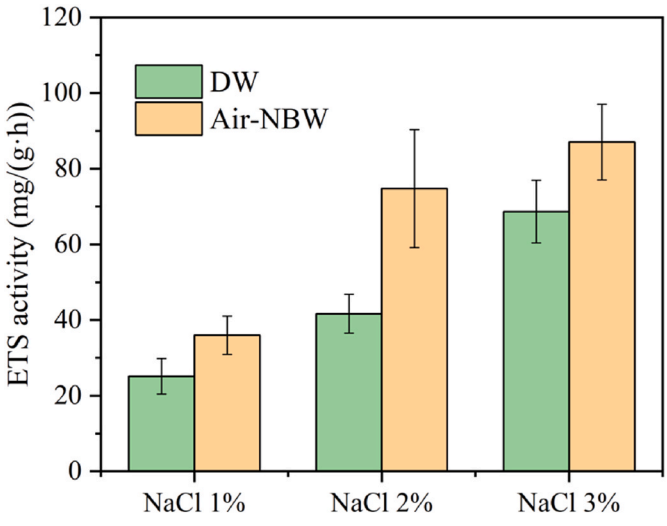


Fig. 2. ETS activity at the end of each phase with an increase in the salinity.

salinity acclimation exhibits potential in significantly reducing the inhibitory impacts of hypersalinity stress in AD systems.

3.2. Effects of inoculum ratio on methanogenesis performance from the FW AD system under a salt level of 3 % using the domesticate inoculum

3.2.1. Methanogenesis performance

The I/S ratio has always been considered a vital parameter affecting the efficiency and stability of AD [27]. Although the effects of different I/S ratios on AD performance have been extensively investigated, their effects on AD of organic solid wastes in adverse environments (such as high salinity) have scarcely been investigated. In this study, the effects of different I/S ratios on methanogenesis performance in the AD of FW under a salt level of 3 % were investigated using the domesticated inoculum, and the relevant results are presented in Fig. 3. Besides, the I/S ratio declined from 2 to 1, and the operating cycle (around 30 d) remained largely unchanged (Fig. 3 A). However, the peak values of 28.61 ± 3.40 and 30.91 ± 0.17 mL/(g-VS_{fed}·d) at the I/S ratio of 1 were identified on 12 and 10 d in the DW and Air-NBW digesters, respectively, appearing earlier than the peak values at the I/S ratio of 2 (on 18 days in both groups).

In contrast, as the I/S ratio declined from 2 to 0.5, the AD digesters were operated approximately for 70 d till almost no biogas production was detected. The Gompertz model was selected to resolve lag-phase dynamics in methane production, as it explicitly parameterizes microbial lag time (λ) under inhibitory conditions. Kinetic analysis revealed that the λ was lengthened with the decline of the I/S ratio, particularly in the DW group (Table 1). To be specific, the λ was extended 2 times in the DW group with the decline of the I/S ratio from 2 to 0.5. As revealed by the above results, the high I/S ratio can lead to the shortened operating period of digesters and the improved methane production rate by supplying more methanogenic bacteria, and other research has drawn similar conclusions [28].

Table 1
Kinetic analysis of CH_4 yield from AD of FW system with 3 % salinity under different I/S ratios.

Digesters	Modified Gompertz model			
I/S ratio = 2				
	P (mL/g-VS_{fed})	R_{max} (mL/(g-VS_{fed}·d))	λ (d)	R^2
DW	193.94 ± 5.82	28.83 ± 1.36	4.00 ± 0.38	0.9957
Air-NBW	225.49 ± 6.73	33.59 ± 1.59	3.96 ± 0.38	0.9957
I/S ratio = 1				
	P (mL/g-VS_{fed})	R_{max} (mL/(g-VS_{fed}·d))	λ (d)	R^2
DW	209.26 ± 4.42	36.35 ± 1.69	5.58 ± 0.34	0.9964
Air-NBW	244.87 ± 4.26	36.99 ± 1.24	3.84 ± 0.27	0.9978
I/S ratio = 0.5				
	P (mL/g-VS_{fed})	R_{max} (mL/(g-VS_{fed}·d))	λ (d)	R^2
DW	321.28 ± 45.17	12.59 ± 0.69	8.33 ± 1.78	0.9556
Air-NBW	363.88 ± 31.61	14.82 ± 0.59	6.25 ± 1.30	0.9752

Fig. 3 B illustrates the corresponding cumulative methane yield in the DW and Air-NBW groups with the different I/S ratios. The cumulative methane yield in the DW groups reached 199.33 ± 9.92 and 262.93 ± 3.69 mL/g-VS_{fed} at I/S ratios of 1 and 0.5, respectively, representing increases of approximately 15 % and 52 % compared to the DW group with an I/S ratio of 2. Compared to the DW groups at equivalent I/S ratios, the Air-NBW digester showed a non-significant increase to 228.88 ± 4.36 mL/g-VS_{fed} at an I/S ratio of 1 ($p = 0.061$) but achieved a statistically significant higher yield of 307.45 ± 3.77 mL/g-VS_{fed} at an I/S ratio of 0.5 ($p = 0.007 < 0.05$). The above results were not consistent with the findings of other studies where the higher methane yield is achieved in the digesters with the high I/S ratio [29,30]. The inconsistency in results may be clarified by differences in the activity of the inoculated microorganisms. As discussed in the research of Elbeshbishy et al. [31], the ability to produce methane could be increased under low

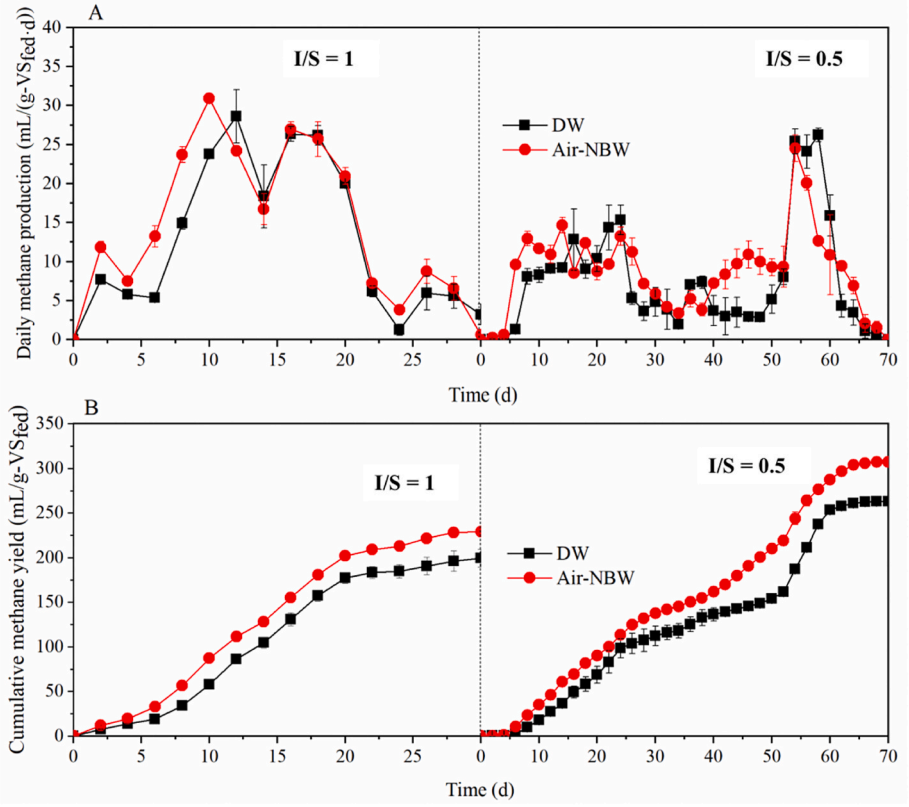


Fig. 3. Effects of I/S ratio (1 and 0.5) on methane production from the FW AD system with 3 % salinity by using the domesticate inoculum. (A) Daily methane production. (B) Cumulative methane yield.

I/S ratios if sufficient nutrients are supplied. Moreover, Kong et al. [32] suggested a higher methane yield at the minimum I/S ratio of 0.25 when inoculating pre-incubated inoculum. In contrast, more inoculum (high I/S ratio) should be provided to metabolize the substrate for respect to non-incubated inoculum with low microbial activity. Besides, Xing et al. [33] confirmed that the cumulative methane production declines decreased the decrease of I/S for the seed sludge without domesticating, whereas the long-term acclimatization of inoculum will change the optimal I/S ratio. In this study, the inoculum was domesticated by Air-NBW, and the gradual increase of the NaCl concentration, such that the sample exhibited high microbial activity and high methane yield. From the perspective of methane production and running time, the I/S ratio of 1 was confirmed to be optimal for AD of FW exhibiting high salinity.

3.2.2. Variations in pH, VFAs, and alkalinity

The pH of the inoculum reached 6.76 ± 0.08 and 6.88 ± 0.03 in the control and Air-NBW digesters, respectively, when the acclimatization process was completed. The original pH of all digesters was fitted to 7.5 in the experiment of this study using 2 M HCl and/or 2 M NaOH solution. As depicted in Fig. 4, the pH fluctuations of all digesters displayed similar trends in the AD process. On that basis, the pH value was first decreased and then increased, and the decrease rate of the pH value was elevated with the reduction of the I/S ratio. Existing research has generally suggested that lower I/S ratios can lead to lower pH values [34], and this finding was also indicated by the result of this study. The minimum pH values in the DW and Air-NBW groups reached 5.93 ± 0.01 and 6.16 ± 0.01 at the I/S ratio of 1 and 5.25 ± 0.00 and 5.39 ± 0.01 at the I/S ratio of 0.5 on 6 d. Moreover, the pH value was recovered to nearly 7.00 (favorable for methanogenesis) on 12 days and 30 days at the I/S ratio of 1 and 0.5, respectively. When the experiment was about to end, the pH values in the DW and Air-NBW groups reached 8.30 ± 0.09 and 8.33 ± 0.02 at the I/S ratio of 1 and 7.49 ± 0.01 and 7.78 ± 0.00 at the I/S ratio of 0.5.

Fig. 4 also presents the changes in TA, such that the pH can always be kept in the ideal range (6.8–7.8) to avoid the accumulation of VFAs in the AD system [35]. At the I/S ratio of 1, the TA continued to increase, whereas the TA was decreased first and then increased in the digesters at the I/S ratio of 0.5. As revealed by the above-mentioned results, low I/S ratios may result in the acidification of the AD digesters, such that methane production can be inhibited, especially in the DW group. In other words, the microorganisms domesticated by Air-NBW turned out to exhibit more resistance to the adverse environment of the system after a long period of salt acclimatization. Furthermore, with the rise in TA, the pH value of all digesters was recovered, and the methanogens can be activated to produce more methane.

The results of pH values corresponded to the variations in VFAs concentration, and the distribution of VFAs from different pathways in the AD process can more significantly indicate the health of AD (e.g., the acidogenesis balance and methanogenesis balance) [36]. Fig. 5 A and B describe the impact of the I/S ratio on VFAs compositions during the AD of FW at the salinity of 3 %. To be specific, the concentration of total VFAs was steadily elevated in all digesters; subsequently, it declined significantly in the digesters at the I/S ratio of 1 but decreased slowly in the digesters with the I/S ratio of 0.5. The maximum concentrations of total VFAs reached 3438.61 ± 53.34 and 3284.36 ± 67.20 mg/L in the control and Air-NBW digesters at the I/S ratio of 1 and 5908.96 ± 41.05 and 6413.28 ± 65.49 mg/L in the control and Air-NBW digesters at the I/S ratio of 0.5, respectively.

The above results may explain why the lag time was longer at the I/S ratio of 0.5, primarily due to the accumulation of VFAs that can lead to the generation of an acidic environment. Thus, methane production will be hindered. For the individual VFA concentration, acetic acid (HAc), and butyric acid predominated in the acid accumulation phase, accounting for nearly 90 % and 88 % at the I/S ratio of 1 and 0.5. Existing research has suggested that HAc is considered a preferable substrate for CH_4 production, such that the high concentration of substrate with the higher methane yield is likely to be explained.

3.2.3. Process stability

The alkalinity and VFAs are considered a buffering combination in AD reactions, and VFA/TA and IA/PA ratios always serve as the greater parameters that are capable of monitoring the stability of the AD system. Fig. 6 presents the profiles of IA/PA ratio and VFA/TA ratio from the FW AD system at the salinity of 3 % using the domesticated inoculum. The IA/PA ratio in the digesters with the I/S ratio of 1 declined steadily from the maximum value (3.61 ± 0.24 in the DW group and 2.51 ± 0.26 in the Air-NBW group) on 6 d to 0.60 ± 0.02 in the DW group and 0.28 ± 0.01 in the Air-NBW group at the end of process, respectively. As revealed by the above result, the digesters at the I/S ratio of 1 exhibited better stability than the digesters at the I/S ratio of 0.5. It is noteworthy that the IA/PA ratio fluctuated considerably in digesters at the I/S of 0.5 in the prophase of the reaction. The above result suggested an extremely unstable system since an abrupt rise in the IA/PA ratio commonly represents a fall in pH buffering ability [16].

Besides, the results of the VFA/TA ratio indicated the process stability. The VFA/TA ratio of digesters with the I/S ratio of 1 dramatically declined, which was lower than 0.4 on 6 days, suggesting that the system regained stability. However, the instability persisted for a long time in the digesters with the I/S of 0.5 in accordance with the higher VFA/TA ratio. The reason for the above result is that the AD process is optimally stable at IA/PA ratio below 0.4, and the digester is prone to significant instability at IA/PA ratio above 0.8, as confirmed by a previous study [37]. Furthermore, the stability of the Air-NBW system was superior to that of the DW system regardless of the I/S ratio.

4. Conclusions

Domestication with Air-NBW significantly reduces start-up time and enhances methane yields in high-salinity digesters. Moreover, the

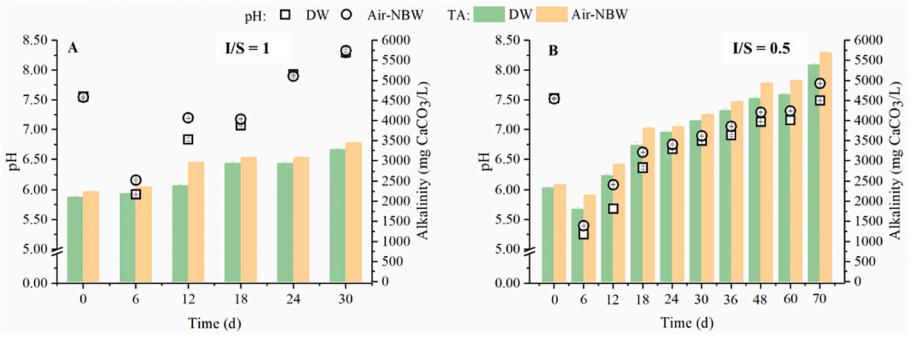


Fig. 4. Variations of pH and alkalinity with the I/S ratio of 1 (A) and 0.5 (B) in the FW AD system with a 3 % salinity system.

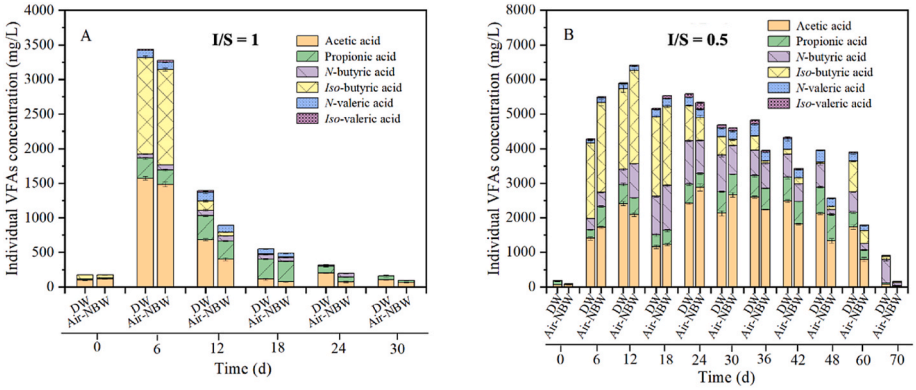


Fig. 5. Impact of I/S ratio (1 (A) and 0.5 (B)) on VFAs compositions and concentrations during the FW AD system with 3 % salinity by using the domesticate inoculum.

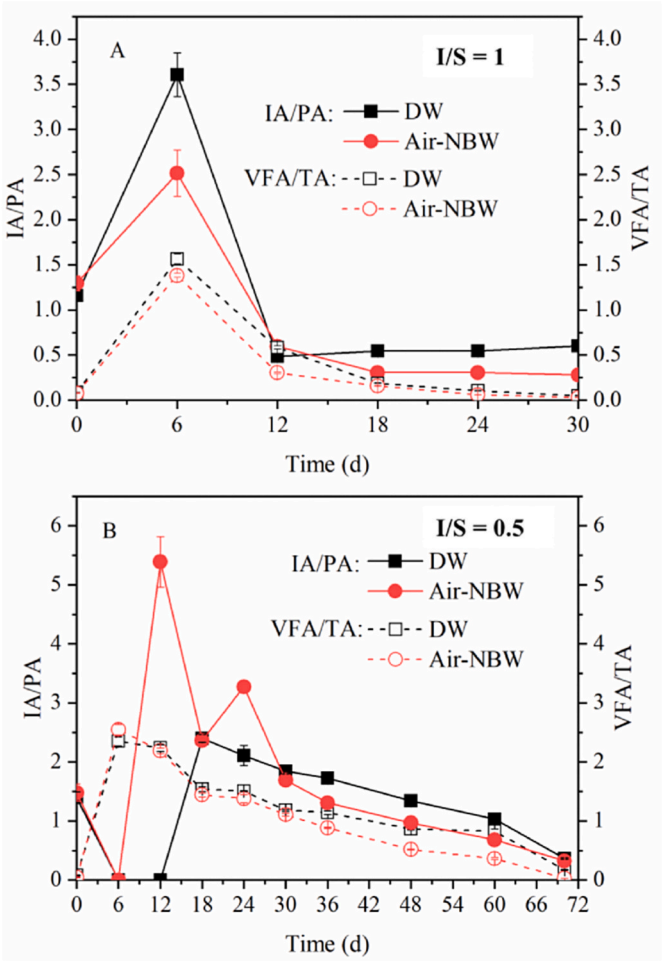


Fig. 6. Profiles of IA/PA and VFA/TA during the FW AD system with 3 % salinity by using the domesticate inoculum. (A) is an I/S ratio of 1, and (B) is an I/S ratio of 0.5.

introduction of Air-NBW and the gradual increase of salt concentration to domesticate microorganisms can be effective in resisting the inhibitory effects of high-salt concentration on the AD system. In general, the I/S ratio of 1 was confirmed to be optimal for AD of FW exhibiting high salinity due to the high methane yield (compared with the I/S ratio of 2) and short operation time and process stability (compared with the I/S ratio of 0.5).

CRediT authorship contribution statement

Tingting Hou: Writing – review & editing, Writing – original draft, Supervision, Project administration, Investigation, Data curation, Conceptualization. **Youzhou Jiao:** Supervision, Methodology. **Jiamin Zhao:** Writing – review & editing. **Zhenya Zhang:** Supervision, Methodology, Conceptualization. **Zhongfang Lei:** Writing – review & editing, Supervision. **Guizhuan Xu:** Writing – review & editing. **Xiaohui Pan:** Writing – review & editing. **Pengfei Li:** Writing – review & editing. **Francesco Petracchini:** Writing – review & editing. **Chao He:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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